

Some definitions

$Subst := set (Var * Exp)$
 $Problem := set Equation$
 $Tuple := set (Subst * Problem)$
 $-\langle _ \rangle : exp \rightarrow Subst \rightarrow exp$
 $dom : Subst \rightarrow set Var$
 $smatch : Tuple \rightarrow Tuple \rightarrow Prop$

smatch steps are classified into three

Congruence steps

$$\begin{aligned} \text{smatch_app_congr} : (s \ t = s' \ t') \in P \rightarrow \\ \text{smatch } (S, P) \ (S, P + (s = s') + (t = t') \setminus (st = s' \ t')) \end{aligned}$$

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σ -min steps

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Instantiation case

$$\begin{aligned} \textit{smatch_inst} : (s = X) \in P \rightarrow \\ X \notin \textit{lhs_vars} \ P \rightarrow \\ S' = S \gg \{(X, s)\} \rightarrow \\ \textit{smatch} \ (S, P) \ (S', (P \setminus (s = X)) \langle \{(X, s)\} \rangle) \end{aligned}$$

Preservation statement

Validity

$$\text{valid_tuple } (S, P) := \underline{(\text{dom } S) \cap (\text{vars_of } P) = \phi}$$

Matching solution

$$\text{match_sol } SI \ (S, P) := \frac{(s = t) \in P \rightarrow s =_{\sigma_{min}} t \langle SI \rangle \wedge \underline{\exists S', S \gg S' =_{\sigma_{min}} SI}}{}$$

Preservation theorem

$$\begin{aligned} \text{match_pres} : & \text{valid_tuple } T \rightarrow \\ & (\text{dom } SI) \cap (\text{lhvars_of } P) \rightarrow \\ & \text{smatch } T \ T' \rightarrow \\ & \text{match_sol } SI \ T' \rightarrow \\ & \text{match_sol } SI \ T \end{aligned}$$